



Switch Gear Rack Mount

Installation Guide

CONNECTING 'SWITCH GEAR RACK' ASSEMBLY

1. Identify Main Breaker
2. CT's should be mounted to the 'Load' side of the Main Breaker
3. Meter should be mounted directly above where CT's are installed due to length of CT's connecting wires.
4. Identify location of 'Switch-Gear' Rack, staying inside of 10 feet from meter.
5. PREP CT's – Tape end of CT connecting wires with phase-colored tape.
6. PREP SWITCH GEAR RACK
 - a. Mount 'L' Brackets at bottom legs of Rack using the ½" nut, bolt and washer.
 - b. Mount 2 XPS600 units to bottom section of rack (Use Spring Nuts).
 - c. Mount 2 Load Controller units to top section of rack making sure to align Load Controller to right-side of XPS600 below (Use Spring Nuts).
7. PREP LEFT-SIDE LOAD CONTROLLER
 - a. Drill ½" hole on the left-side of Load Controller, towards the front and in between internal components for later connecting VT's wires to Meter.
8. PREP METER
 - a. Drill ½" hole on right-side of Meter case, towards the front and in between internal components for later connecting the other end of VT wires from left-side Load Controller.
 - b. Drill 1 1/8" hole on right-side of Meter case, towards the front and in between internal components for later connecting the CT wires (coming out of Switch Gear).
9. PREP SWITCH GEAR
 - a. Using 2 people and a latter on each side, lift and position the Switch Gear Rack onto the top-side of the Switch Gear where the holes will be drilled, also staying inside of 10 feet from where Meter will be installed.
 - b. Once Switch Gear Rack is positioned, use a Sharpie to mark the 4 of the ½" holes for drilling.
 - c. Then, mark two (2) - 1 1/8" holes for the 6-gauge connecting wires from both 'Load Controllers' to the independent 'Breaker' located on the Switch Gear. IMPORTANT NOTE: It is best to locate these holes on the left-side back or middle back of the Switch Gear Rack.
 - d. And lastly, mark a third 1 1/8" hole near the Meter for running the CT wires into the Meter.
10. PREPARE FOR DRILLING SWITCH GEAR

IMPORTANT NOTICE: BEFORE DRILLING THE HOLES IN THE SWITCH GEAR, MAKE SURE THAT PROPER PPE SAFETY GEAR IS WORN AND MAKE SURE TO PROTECT THE INSIDE OF THE SWITCH GEAR WITH NON-CONDUCTIVE MATERIAL (i.e. Box or similar) TO ENSURE THAT NO METAL SHAVINGS ENTER THE SWITCH GEAR WHILE DRILLING.

- a. Re-position (out-of-the way) the Switch Gear Rack to begin drilling both ½' and 1 1/8" holes for mounting.
- b. Once all holes are drilled, re-position the Switch Gear Rack for ½" screw mounting. Remember that the 1 1/8" holes will later be used for connecting Load Controllers to Switch Gear Breaker.
- c. Once Switch Gear Rack is securely mounted, install a ¾" Coupler into each of the 1 1/8" holes to be used with the 'Seal-Tite' or 'flex' conduit. (2 – for Load Controllers and 1 – for Meter).

11. CONNECTING 6-GAUGE WIRES FROM LOAD CONTROLLERS TO SWITCH GEAR BREAKER

- a. Measure length of 6-gauge wire from each Phase of each Load Controller to Switch Gear Breaker making sure to leave an additional 2 to 3 feet of wire.
- b. Once all 6 wires are sized and cut, then use color tape on both ends to indicate Phase before connecting.
- c. Then, go to 'Load Controller' to size and cut 'flex' conduit from each 'Load Controller' to 'coupler' located on the 'Switch Gear'. Run 6-gauge wire from inside 'Switch Gear', through 'flex' conduit to 'Load Controller'. Make connection to 'Breaker' inside 'Load Controller' along with 14-gauge wire from 'Meter'.
- d. Tie-wrap 6-gauge wire from inside of 'Switch Gear' to 'Breaker' located on the 'Switch Gear'.
- e. Make connection of 6-gauge wire to 'Breaker' in the 'Switch Gear'.

12. CONNECTING CT'S TO METER

- a. Attached each CT to each phase (on the 'Load side'). Make sure arrows located on each CT is pointed in the direction of the 'load'. NOTE: Make sure exposed wires at the other end of each CT is not exposed to ANY part of the inside of the 'Switch Gear', until ready to make connection to 'Meter'.
- b. Then, take other end of each CT and run through the 1 1/8" hole and 'coupler' for the 'Meter'.
- c. Size 'flex' conduit from 'Meter coupler' to 'Meter' case.
- d. Run CT wires through 'flex' conduit and then through the 'coupler' on the 'Meter'.
- e. Then, screw each CT wire (white, brown and ground) to the green connector located on the 'Meter'.
- f. Insure that 'Meter' System Parameters are properly set for the 'Meter' before turning 'ON'. To install proper system parameters for 'Meter', take the USB cable end (Pi side) and plug into a USB slot on your computer. Then, go into 'Viewpoint' software and upload system parameter file directly into 'Meter'. (**See "Installing System Parameters onto Meter"**)
- g. Once all wires are connected, go to 'Load Controller' and turn 'ON' the 60 amp dinrail 'Breaker' inside the 'Load Controller' case.

BEFORE TURNING 'ON' BREAKER AT THE 'SWITCH GEAR', DOUBLE CHECK ALL WIRE CONNECTIONS.

13. TURNING 'ON' BREAKER AT THE 'SWITCH GEAR'.

- a. Once all wire connections have been confirmed, turn 'ON' breaker.
- b. Then, go to 'Meter' case to check the LED indicators on 'Meter' to insure proper phase connections. If correct, the all LED's for each phase of the 'Meter' will indicate 'GREEN'.
(See 'Correcting Meter Polarity')
- c. IF NOT, 'go to trouble-shooting section' for correcting 'Meter' polarity.

(Training Required) - INSTALLING SYSTEM PARAMETERS ONTO METER.

1. First, install 'ViewPoint' software onto your PC.
2. Connect USB cable from the Meter to the USB port on your computer.
3. Double-Click on Viewpoint icon
4. Click 'CONNECT' button
5. Upload the Meter Parameter file into the Meter.